

# MAE 5036: Advanced Computational Solid Mechanics

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| <b>Course Level:</b> | Graduate (First-Year MS/PhD)                                                                                |
| <b>Credits:</b>      | 3 (average 3 hours per week)                                                                                |
| <b>Meeting Time:</b> | Every Wednesday, 7pm – 10 pm (approximately 3 × 50 minutes per week), 224 1 <sup>st</sup> teaching building |
| <b>Instructor:</b>   | Jiarui Wang , 819 North Engineering Building,<br>wangjr@sustech.edu.cn                                      |

## Course Description

Topics include tensor analysis, kinematics of deformation, stress measures, balance laws, constitutive theory, and linear and finite elasticity. The course establishes the theoretical foundations for advanced studies in nonlinear solid mechanics and computational mechanics.

## Course Objectives

1. Use tensor notation and tensor calculus in continuum formulations.
2. Describe finite deformation using appropriate kinematic measures.
3. Formulate stress tensors and interpret their physical meaning.
4. Derive conservation and balance laws in local form.
5. Apply principles of material frame indifference and isotropy.
6. Develop constitutive models for linear and hyperelastic materials.
7. Analyze simple boundary value problems in both small and finite deformation settings.

## Weekly Schedule

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| <b>Section 1</b> | <b>Introduction to Nonlinear Finite Element Methods (Week 1-3)</b> <ul style="list-style-type: none"><li>• Review of linear finite element methods</li><li>• Sources of nonlinearity</li><li>• Newton–Raphson method for one-dimensional problems</li></ul> <b>非线性有限元方法简介（第 1 – 3 周）</b> <ul style="list-style-type: none"><li>• 线性有限元方法回顾</li><li>• 非线性的来源</li><li>• 一维问题中的 Newton – Raphson 迭代方法</li></ul> |
| <b>Section 2</b> | <b>Nonlinear Statics (Week 4-6)</b> <ul style="list-style-type: none"><li>• Small-deformation nonlinear elasticity</li><li>• Weak form and linearized operators</li><li>• Finite element formulation</li></ul>                                                                                                                                                                                               |

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|           | <ul style="list-style-type: none"> <li>• Basic implementation and code development</li> </ul> <p>非线性静力学问题的有限元方法（第 4 - 6 周）</p> <ul style="list-style-type: none"> <li>• 小变形非线性弹性</li> <li>• 弱形式与线性化算子</li> <li>• 有限元离散与数值格式</li> <li>• 基本程序实现与代码开发</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| Section 3 | <p><b>Solution Methods for Nonlinear Algebraic Systems (Week 7-8)</b></p> <ul style="list-style-type: none"> <li>• Newton and modified Newton methods</li> <li>• Consistent linearization</li> <li>• Line search methods</li> <li>• Arc-length methods</li> </ul> <p>非线性方程组的求解方法（第 7 - 8 周）</p> <ul style="list-style-type: none"> <li>• Newton 法与修正 Newton 法</li> <li>• 一致线性化</li> <li>• Line search</li> <li>• Arc-length method</li> </ul>                                                                                                                                                     |
| Section 4 | <p><b>Finite Deformation and Hyperelasticity (Week 9-11)</b></p> <ul style="list-style-type: none"> <li>• Continuum mechanics background</li> <li>• Weighted residual and variational formulations</li> <li>• Total and updated Lagrangian formulations</li> <li>• Finite element discretization</li> <li>• Material and geometric stiffness</li> <li>• Code development</li> </ul> <p>大变形与超弹性问题的有限元方法（第 9 - 11 周）</p> <ul style="list-style-type: none"> <li>• 连续介质力学基础</li> <li>• 加权余量形式与变分形式</li> <li>• 全拉格朗日格式与更新拉格朗日格式</li> <li>• 有限元离散</li> <li>• 材料刚度与几何刚度</li> <li>• 程序实现与代码开发</li> </ul> |
| Section 5 | <p><b>Small- and Finite-Strain Elastodynamics (Week 12-14)</b></p> <ul style="list-style-type: none"> <li>• Semi-discrete weighted residual formulations</li> <li>• Classical time integration algorithms</li> </ul>                                                                                                                                                                                                                                                                                                                                                                              |

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|           | <ul style="list-style-type: none"> <li>• Stability, consistency, and convergence</li> <li>• Nonlinear solution algorithms for dynamics</li> <li>• Basic finite element implementation</li> </ul> <p>动力问题的有限元方法（第 12 – 14 周）</p> <ul style="list-style-type: none"> <li>• 半离散加权余量形式</li> <li>• 时间积分算法</li> <li>• 稳定性、相容性与收敛性分析</li> <li>• 非线性动力学问题的求解算法</li> <li>• 基本有限元程序实现</li> </ul>                                                                                                                                                                                                                                                     |
| Section 6 | <p><b>Advanced Numerical Method in Computational Solid Mechanics (Week 15-16)</b></p> <ul style="list-style-type: none"> <li>• Least-squares approximations, kernel approximations, derivative approximations</li> <li>• Smoothed particle hydrodynamics</li> <li>• Element-free Galerkin method</li> <li>• Reproducing kernel particle method</li> <li>• Imposing boundary conditions</li> <li>• Domain integration</li> </ul> <p>计算固体力学中的无网格类方法（第 15 – 16 周）</p> <ul style="list-style-type: none"> <li>• 最小二乘近似、核近似与导数近似</li> <li>• 光滑粒子流体动力学方法</li> <li>• 无单元伽辽金法</li> <li>• 再生核粒子法</li> <li>• 边界条件施加方法</li> <li>• 弱形式的积分方法</li> </ul> |

### Grading Policy

- Attendance: 10%
- Homework: 35%
- Midterm Exam: 15%
- Final Project and oral presentation: 40%